

Dietary microparticles and their impact on tolerance and immune responsiveness of the gastrointestinal tract

Dietary microparticles are non-biological bacterial-sized particles of the gastrointestinal lumen that occur due to endogenous formation (calcium phosphate) or following oral exposure (exogenous microparticle). In the UK, about 40 mg (1012) of exogenous microparticles are ingested per person per day, through exposure to food additives, pharmaceutical/supplement excipients or toothpaste constituents. Once ingested, exogenous microparticles are unlikely to pass through the gastrointestinal tract without adsorbing to their surfaces some ions and molecules of the intestinal lumen. Both entropy and ionic attraction drive such interactions. Calcium ions are especially well adsorbed by dietary microparticles which then provide a positively charged surface for the attraction (adsorption) of other organic molecules such as lipopolysaccharides, peptidoglycans or protein antigen from the diet or commensal flora. The major (but not only) sites of microparticle entry into intestinal tissue are the M-cell rich lymphoid aggregates (termed Peyer's patches in the small bowel). Indeed, it is well established that this is an efficient transport route for non-biological microparticles although it is unclear why. We hypothesise that this pathway exists for "endogenous microparticles" of calcium phosphate, with immunological and physiological benefit, and that "exogenous dietary microparticles", such as titanium dioxide and the silicates, hijack this route. This overview focuses on what is known of these microparticles and outlines their potential role in immune tolerance of the gut (endogenous microparticles) or immune activation (exogenous microparticles) and inflammation of the gut.